4.3 Applications in Physics Problems

4.3.1 Magnetic Phase Transition

The Ising model explains the ferromagnetic phase transition.[3] Using the mean field approximation, the magnetization per atom satisfies the following transcendental equation

$$m = \mu \tanh\left(\frac{Cm}{k_{\mathrm{B}}T}\right) \tag{4.23}$$

where C is a positive constant determined by the spin interaction between atoms, T is temperature, and μ is the magnetic moment each spin carries. We want to know the magnetization as a function of temperature by solving Eq. (4.23). By direct inspection $m = 0$ is always a solution. However, depending on temperature, there are more solutions. We want to plot all roots as functions of temperature. First, we simplify the expression by introducing normalized magnetization $\tilde{m} = m/\mu$ and normalized temperature $\tilde{T} = \frac{k_{\mathrm{B}}T}{\mu C}$. The normalized quantities are dimensionless and various constants disappear. We omit tilde in the normalized quantities to avoid cluttering. Then, we solve

$$f(m) \equiv m - \tanh(m/T) = 0 \tag{4.24}$$

utilizing its derivative

$$f'(m) = 1 - \frac{\operatorname{sech}^2(m/T)}{T} \tag{4.25}$$

To use the Newton-Raphson method, we need to start with a good initial guess. As Fig 4.5 shows, $\tanh(m) \to 1$ for $m \gg 1$. Therefore, the root must be below $m = 1$. As T approaches zero, the root approaches one. Therefore, $m = 1$ is a good starting point for low temperature case. As temperature increases, the root gradually decreases toward zero. Therefore, when temperature is lowered a little bit, the previous root can be a very good starting point of the next root finding process. We don't have to bracket the root in this case. The results are shown in Fig. 4.5b.

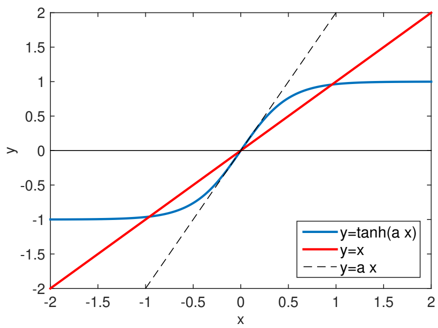
EXAMPLE 4.5 Bifurcation of magnetization

Program 17.2 finds the roots of Eq. (4.23) by Newton-Raphson method and plots them. The result is shown in Fig. 4.5b. Above the critical temperature $S_{\mathrm{C}} = 1$, there is only one root and the magnetization vanishes. Below S_{C}, positive and negative magnetization spontaneously appear. The root at $m = 0$ still exists but it is no thermodynamically not stable.

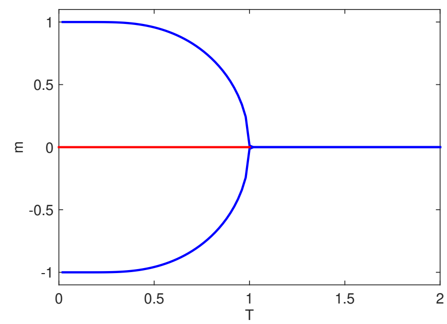
Exercise 4.5 Spontaneous vs induced magnetization
When external magnetic field H is applied, the mean-field equation becomes

$$m = \mu \tanh\left(\frac{\mu H + Cm}{k_{\mathrm{B}}T}\right) . \tag{4.26}$$

Modify Program 17.2 and plot the magnetization as a function of temperature with a fixed value of $H \neq 0$. Try various different values of H. How is the spontaneous magnetization shown in Fig. 4.5b modified by the magnetic field? [Use the normalized magnetic field $\tilde{H} = H/C$.]



(a) The magnetization is determiend by the crossing of $y = x$ and $y = \tanh(ax)$ where $a = 1/T$. When $a < 1$ $x = 0$ is only the solution. For $a > 1$, two more solutions appear. $a = 1$ corresponds to the critical point of the ferromagnetic transition.



(b) Bifurcation of magnetization. For $T > 1$ there is only one magnetic state $m = 0$. When temperature is lowered below a critical temperature $T_{\mathrm{C}} = 1$, two stable states, one with positive m and the other with negative m. $m = 0$ is still a solution but unstable one. At $T = 0$, the magnetization reaches the maximum possible value $m = \pm 1$ where all electrons have the same spin state.

Figure 4.5: Ferromagnetic Phase Transition

4.3.2 Energy of a Quantum Particle in a Square Potential

A particle of mass m is trapped in a square well potential of depth V_0 and width $2a$. Energy eigenvalue E is determined by Schödinger equation

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right]\psi(x) = E\psi(x) \tag{4.27}$$

Analytical calculation ends up with the following transcendental equations[4]

$$z\tan(z) - \sqrt{z_0^2 - z^2} = 0 \qquad \text{for symmetric states} \tag{4.28a}$$

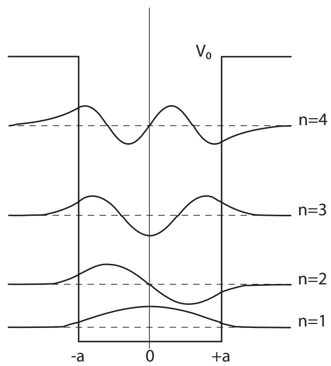
$$z\cot(z) + \sqrt{z_0^2 - z^2} = 0 \qquad \text{for anti-symmetric states} \tag{4.28b}$$

where

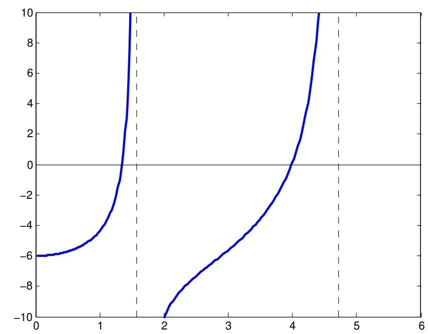
$$z = \sqrt{\frac{2ma^2E}{\hbar^2}} \tag{4.29a}$$

$$z_0 = \sqrt{\frac{2ma^2V_0}{\hbar^2}} \tag{4.29b}$$

The roots of Eqs (4.28) determines the energy eigenvalues. WE want to know all eigenvalues. However, the number of roots depends on the system parameters, which makes it difficult to bracket a root. We need to inspect the location of the roots. In Fig. the left hand side of Eq (4.28a) is plotted. First of all, we note that the function is defined for $z \in (0, z_0)$ and thus all roots must be between 0 and z_0. Considering $\tan x$ diverges at $2\pi(n \pm 1/4)$, each root must be between the diverging points. Hence, we bracket the roots in $(0, \frac{\pi}{2} - \delta)$, $(\frac{\pi}{2} + \delta, \frac{3\pi}{2} - \delta)$, $\cdots$, $2\pi(n + \frac{1}{4}) + \delta, z_0)$. Here δ is a small positive quantity to avoid the singularities. Program



(a) A square well potential with depth V_0 and width $2a$. Two symmetric and two anti-symmetric energy eigenstates are plotted.



(b) The left hand side of Eq (4.28a) is plotted for $z_0 = 6$. There are two roots. Clearly the root is bracketed by the dashed lines at $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$.

Figure 4.6: Quantum particle in a finite suare well

4.6 finds all roots for given z_0. Ten iterations of the bisection methods followed by the secant method find two roots.

```
root=1.34475105
root=3.98582621
```

4.3.3 Classical Turning Points

In Section 3.6.1, we tried to calculate the period of oscillation but we were not quite ready at that time because we did not know how to find the turning points. Now we can find them using a numerical root finding method. Suppose that a particle is bound in a potential

$$U(x) = U_0 \left[\sin\left(\frac{2\pi x}{L}\right) - \frac{1}{4}\sin\left(\frac{4\pi x}{L}.\right)\right] \tag{4.30}$$

We want to find the period of the bound motion using the formula (3.26). The turning points are the roots of $U(x) = E$. The potential is simple and we can find the analytic expression of its derivative. Hence, the Newton-Raphson method can be used to find the roots.

Exercise 4.6 Modify Program 4.4 and find a pair of the classical turning points of the potential (18.39). Using the results, find the period of oscillation.

4.3.4 Closest Approach in Scattering

In Section 3.6.2, we discussed how to evaluate the scattering angle (3.35). In order to carry out the integral, we still have to find the closest approach r_0 determined by Eq. (3.36). For the screened Coulomb potential, we are able to calculate the first order derivative of the equation. The Newton-Raphson method seems a good choice.

Exercise 4.7 Solve Eq. (3.36) for the Yukawa potential. Then, calculate the scattering angle, Eq. (3.35), using the method discussed in Section 3.6.2.

4.4 Problems

4.1 The interaction energy between two atoms is often described by the Morse potential[5]

$$U(r) = D_e \left[\mathrm{e}^{-2(r-r_0)/a} - 2\mathrm{e}^{-\alpha(r-r_0)/a} \right] \tag{4.31}$$

where r is the distance between the atoms. The parameters, D_e. r_0, and a are, dissociation energy, equilibrium distance, and the range of interaction, respectively. To simplify numerical calculation, we normalize energy and distance by D_e and a. In addition, we use the displacement from the equilibrium position as the coordinate. Then, the potential is simply

$$U(x) = \mathrm{e}^{-2x} - 2\mathrm{e}^{-x} . \tag{4.32}$$

were $x = (r - r_0)/a$. Find the classical turning points for $E = -0.5$ (This means the actual energy is $E = -D_e/2$ since we use D_e as the unit of energy.) Do not forget to plot the potential and visually inspect the turning point before using numerical root finding methods.

4.2 Find the eigenvalue of all asymmetric state by solving Eq. (4.28b) for $z_0 = 6$.

4.3 A quantum ball of mass m is bouncing on a hard floor under a uniform gravity g.[6] Schrödinger equation for the particle is

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi(y)}{\mathrm{d}y^2} + mgy\,\psi(y) = E\psi(y). \tag{4.33}$$

The wavefunction ψ must vanish at the floor (boundary condition). To find the energy eigenvalue, we first simplify the Schrödinger equation. Introducing a new variable $z = \alpha(y - E/mg)$ where

$$\alpha = \left(\frac{2m^2 g}{\hbar^2} \right)^{1/3} \tag{4.34}$$

Eq. (4.33) is rewritten as

$$\frac{\mathrm{d}^2\psi(z)}{\mathrm{d}z^2} = z\psi(z) \tag{4.35}$$

This equation is known as Airy equation and its solution is Airy function $\mathrm{Ai}(z)$. Hence, the energy eigenfunction is $\psi(z) = N\,\mathrm{Ai}(z)$ where N is a normalization constant. Now, we apply the boundary condition. At $y = 0$. $z = -\alpha E/mg$ and thus $\mathrm{Ai}(-\alpha E/mg) = 0$. This boundary condition indicates that the energy E is determined by the roots of the Airy function. Denoting the n-th root as z_n, the energy is $E_n = -mgz_n/\alpha$. The bouncing quantum ball has discrete energy! Find the smallest root of the Airy function and corresponding energy E_1. Use the built-in Airy function `airy()` in MATLAB. For C or Fortran, GNU Scientific Library (GSL)* provides Airy function.

*You can find GSL at `https://www.gnu.org/software/gsl/`.